B Bharath Vishnu

9947269031 | bharathvb1204@gmail.com | Linkedin | Github

EDUCATION

TKM College of Engineering Nov. 2021 – June 2025
Bachelor of Technology in Computer Science Engineering Kollam, Kerala

EXPERIENCE

Tata Consultancy Services July. 2025 – Present
System Engineer Bangalore, IN

- Engineered reusable and modular React.js components, accelerating feature development and improving maintainability of enterprise banking applications.
- Developed scalable Spring Boot microservices and REST APIs handling high-volume transaction and data-processing workloads.

PROJECTS

Task Queue System

- Developed a task queue system using **Node.js** and **Redis** for asynchronous job processing.
- Designed and implemented **RESTful APIs** for job submission, monitoring, and status tracking.
- Optimized the system for high concurrency and parallel task execution.
- Built scalable worker services for background processing and compute-heavy tasks.
- Developed a **React.js** dashboard to monitor queue metrics and job status in real time.

Distributed Log Ingestion & Detection System

- Designed and developed a distributed log ingestion system using **Java Spring Boot** and to process high-volume event streams.
- Built **RESTful APIs** for log collection, parsing, normalization, querying, and analysis.
- Integrated **PostgreSQL** for efficient storage and fast search capabilities.
- Implemented a Python-based anomaly detection module for identifying suspicious log patterns.
- Developed a **React.js** frontend for visualization and monitoring.

Inventory Management System

- Developed an inventory management system using **Java Spring Boot** for managing products, stock levels, and transactions.
- Implemented pagination, filtering, and validation for efficient querying and data consistency.
- Integrated **Redis caching**, significantly improving retrieval performance over direct database queries.
- Built a **React.js** frontend for inventory visualization and management.
- Deployed the application using **Docker Compose** (App + DB + Cache), reducing local setup time by 90%.

Underwater Image Enhancement Model

- Implemented a **GAN-based** underwater image enhancement model using **TensorFlow** trained on the **UIEB** dataset.
- Improved underwater image clarity, color correction, and contrast enhancement.
- Achieved approximately **35% improvement** in PSNR and SSIM benchmark metrics.
- Optimized preprocessing and inference pipelines, reducing inference latency by nearly **50%**.
- Packaged training and inference workflows into reusable Python scripts for reproducibility.

SKILLS

Languages : C# (.Net Core)Java (Springboot), Python (FastAPI), JavaScript (React.js, Next.js, Node.js), SQL (PostgreSQL, MySQL)
Tools and Frameworks : Flutter, Firebase, Docker, GitHub, Postman